Applications of Higher Order Partial Derivatives

Trim Ch 11, S. 12.5

https://www.thestrad.com/video/violin-string-vibrations-captured-on-video/5897.article

Outline:

1. The Laplace Equation
2. The Wave Equation

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The Laplace Equation

The three-dimensional Laplace Equation is given by,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

The Laplace Equation can also be used in two-dimensions, which is given as,

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

Solutions of this equation are called **harmonic functions**.

Harmonic functions play a role in problems of heat conduction, fluid flow and electrostatic potential (see examples 12.12 and 12.13 (page 815), Trim Textbook).

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The One-Dimensional Wave Equation

The wave equation describes, among other applications, the vibration of a violin string.

It is given by,

$$\frac{\partial^2 \omega}{\partial t^2} = c^2 \frac{\partial^2 \omega}{\partial x^2}$$

where,

ω: wave height
x: distance, starting at one end of string
t: time
c: velocity at which wave propagates

Note the regular pattern of peaks and valleys in an instant of time.

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Application Problem – Midterm Level

If f and g are any twice-differentiable functions of one variable, show that,

$$\omega = f(\underbrace{x - ct}_{u}) + g(\underbrace{x + ct}_{v})$$

Satisfies the one-dimensional wave equation:

$$\frac{\partial^2 \omega}{\partial t^2} = c^2 \frac{\partial^2 \omega}{\partial x^2}$$

Solution:

To show that ω satisfies the one-dimensional wave equation, we need to demonstrate that the left-hand side (LHS) is equal to the right-hand side (RHS) of the wave equation, using differentiation.

Application Problem – Midterm Level

Because we have a composition of functions, let us do the following change of variables,

$$\left.\begin{array}{l} \rightarrow u = x - ct \\ v = x + ct \end{array}\right\} \quad w = f(u) + g(v)$$

LHS:

$$\frac{\partial^2 w}{\partial t^2} = \frac{\partial}{\partial t}\underbrace{\left(\frac{\partial w}{\partial t}\right)}_{\mathrm{I}}$$

$$\mathrm{I} \quad \frac{\partial w}{\partial t} = \frac{\partial f(u)}{\partial t} + \frac{\partial g(v)}{\partial t}$$

Note:

The chain rule of one variable stays that if $y = f(u)$ and $u = u(t)$,

$$\frac{df(u)}{dt} = \frac{df(u)}{du} \cdot \frac{du(t)}{dt}$$

Application Problem – Midterm Level

APPLYING CHAIN RULE,

$$\frac{\partial w}{\partial t} = \frac{\partial f(u)}{\partial u}\cdot\underbrace{\frac{\partial u}{\partial t}}_{-c} + \frac{\partial g(v)}{\partial v}\cdot\underbrace{\frac{\partial v}{\partial t}}_{+c}$$

$$\frac{\partial w}{\partial t} = -c\,\frac{\partial f(u)}{\partial u} + c\,\frac{\partial g(v)}{\partial v} \quad --- \quad \mathrm{I}$$

BUT WE NEED $\frac{\partial^2 w}{\partial t^2}$, USING I

$$\frac{\partial}{\partial t}\left(\frac{\partial w}{\partial t}\right) = -c\,\frac{\partial}{\partial t}\left[\frac{\partial f(u)}{\partial u}\right] + c\,\frac{\partial}{\partial t}\left[\frac{\partial g(v)}{\partial v}\right]$$

$$= -c\,\frac{\partial}{\partial u}\left[\frac{\partial f(u)}{\partial u}\right]\cdot\underbrace{\frac{\partial u}{\partial t}}_{-c} + c\,\frac{\partial}{\partial v}\left[\frac{\partial g(v)}{\partial v}\right]\cdot\underbrace{\frac{\partial v}{\partial t}}_{+c}$$

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Application Problem – Midterm Level

$$\frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 f(u)}{\partial u^2} + c^2 \frac{\partial^2 g(v)}{\partial v^2} \quad \ldots \text{ LHS}$$

RHS: $$\frac{\partial^2 w}{\partial x^2} = \frac{\partial}{\partial x}\Big(\underbrace{\frac{\partial w}{\partial x}}_{\text{II}}\Big)$$

$$\frac{\partial w}{\partial x} = \frac{\partial f(u)}{\partial x} + \frac{\partial g(v)}{\partial x}$$

$$\frac{\partial w}{\partial x} = \frac{\partial f(u)}{\partial u} \cdot \underbrace{\frac{\partial u}{\partial x}}_{=1} + \frac{\partial g(v)}{\partial v} \cdot \underbrace{\frac{\partial v}{\partial x}}_{=1}$$

$$\frac{\partial w}{\partial x} = \frac{\partial f(u)}{\partial u} + \frac{\partial g(v)}{\partial v}$$

Application Problem – Midterm Level

APPLYING CHAIN RULE AGAIN,

$$\frac{\partial^2 w}{\partial x^2} = \frac{\partial}{\partial u}\left[\frac{\partial f(u)}{\partial u}\right] \cdot \underbrace{\frac{\partial u}{\partial x}}_{=1} + \frac{\partial}{\partial v}\left[\frac{\partial g(v)}{\partial v}\right] \cdot \underbrace{\frac{\partial v}{\partial x}}_{=1}$$

$$\frac{\partial^2 w}{\partial x^2} = \frac{\partial^2 f(u)}{\partial u^2} + \frac{\partial^2 g(v)}{\partial v^2} \quad \text{--- RHS.}$$

MULTIPLYING LHS BY c^2,

$$\frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial x^2}$$

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Summary

- Higher order partial derivatives appear in many physical problems in the form of partial differential equations (PDEs)
- Applications include,
  - The Laplace Equation (e.g. heat conduction, fluid flow, electric potential)
  - The wave equation (e.g. vibration of strings)
- The chain rule is a very useful tool to solve PDEs but could become cumbersome as the number of variable increases

Thank you